| Question | Answer | Marks | Partial Marks |
|---|---|---|---|
| 2(c) | 15 380 | **4** | **M3** for (1 120 000 – 5000) ÷ (70 + 2.5) oe<br>or<br>**B2** for answer figs 15 379 to figs 15 380<br>or<br>**M2** for (1 120 000 ± 5000) ÷ (70 ± 2.5) oe<br>or<br>**M1** for one of figs 675, 725, 1115, 1125 seen |
| 2(d) | 1.8[0] or 1.801 to 1.802 [million] nfww | **2** | **M1** for figs $16 \times \left(1+\frac{2.4}{100}\right)^5$ oe |
| 3(a) | Correct box-and-whisker plot | **4** | **B1** for lowest value and highest value at 30 and 90<br>**B1** for LQ and UQ at 50 and 72<br>**B1** for median at 63 |
| 3(b)(i) | 56 | **2** | **M1** for 24 soi |
| 3(b)(ii) | 16 | **2** | **B1** for 64 written |
| 3(c)(i) | 14, 22 | **1** | |
| 3(c)(ii) | 61.5 | **4** | **M1** for 35, 45, 55, 65, 75, 85 soi<br><br>**M1** for $\Sigma fx$<br><br>**M1 dep** for *their* $\Sigma fx$ ÷ (8 + 12 +*their* 14 + *their* 22 + 14 +10) or $\Sigma fx$ ÷ 80 |
| 3(c)(iii) | $\frac{35}{69}$ oe | **3** | **M2** for [2]$\left(\frac{10}{24}\times\frac{14}{23}\right)$ oe<br><br>or **M1** for $\frac{10}{24}$ or $\frac{14}{24}$ oe seen<br>If 0 scored, **SC1** for answer $\frac{35}{72}$ oe |
| 4(a)(i) | $\frac{10}{3}$ or $3\frac{1}{3}$ or 3.33[3…] | **3** | **M1** for $42 - 12x = 3x - 8$ oe<br>or for $7 - 2x = \frac{3x}{6} - \frac{8}{6}$ oe<br>**M1** for reaching $ax = b$ correctly FT *their* first step |
| 4(a)(ii) | –2.5 or $-2\frac{1}{2}$ or $-\frac{5}{2}$ | **3** | **M1** for $3 \times 2x = 2(x - 5)$ oe<br><br>**M1** for reaching $ax = b$ correctly FT *their* first step |

| Question | Answer | Marks | Partial Marks |
|---|---|---|---|
| 4(b)(i) | $2(x + 12y)(x – 12y)$ final answer | **3** | **B2** for $(2x + 24y)(x – 12y)$<br>or $(2x – 24y)(x + 12y)$<br>or for $2(x + 12y)(x – 12y)$ seen<br>OR<br>**M2** for $k(x + 12y)(x – 12y)$<br>or **M1** for $2(x^2 – 144y^2)$ |
| 4(b)(ii) | $(5x – 8)(x + 5)$ final answer | **2** | **M1** for $5x(x + 5) – 8(x + 5)$<br>or $x(5x – 8)+ 5(5x – 8)$<br>or for $(5x + a)(x + b)$ where $ab = – 40$ or $a + 5b = 17$ |
| 4(c) | $4x^2 – 17x + 9$ [= 0] oe | **B1** | |
| | $\frac{[--]17 \pm \sqrt{([-]17)^2 - 4(4)(9)}}{2\times 4}$ | **B2** | **FT** *their* 3 term quadratic<br>**B1FT** for $\sqrt{([-]17)^2 - 4(4)(9)}$ or better<br>or $\left(x - \frac{17}{8}\right)^2$ oe or $\sqrt{\frac{([-]17)^2 - 4(4)(9)}{4}}$ or better<br>and **B1FT** for $\frac{[--]17 + \sqrt{q}}{2(4)}$ or $\frac{[--]17 - \sqrt{q}}{2(4)}$ or better<br>or $\frac{17}{8} + \sqrt{\frac{145}{64}}$ oe or $\frac{17}{8} - \sqrt{\frac{145}{64}}$ oe or $\frac{\frac{[--]17}{2} + \sqrt{q}}{4}$ or $\frac{\frac{[--]17}{2} - \sqrt{q}}{4}$ |
| | 0.62 and 3.63 cao | **B2** | **B1** for each<br>**SC1** for 0.6[0] or 0.619 to 0.620 **and** 3.6[0] or 3.6301 to 3.6302<br>or 0.62 and 3.63 seen in working<br>or –0.62 and–3.63 as final answers |
| 5(a)(i) | 62<br>and<br>Angle at centre is twice angle at circumference oe | **2** | **B1** for either |
| 5(a)(ii) | 117<br>and<br>Isosceles [triangle]<br><br>and<br>Opposite angles in a cyclic quadrilateral are supplementary | **4** | **B2** for 117<br>or **B1** for [angle *OCD* =] 28<br>**B1dep** for isosceles [triangle]<br><br>and<br>**B1** for opposite angles in a cyclic quadrilateral are supplementary |